Nonparametric Predictive Regression with Exogenous Shocks: A Sieve Empirical Likelihood Approach (In progress)

Siyan Dong

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1 Introduction and Data-Generating Process

We consider the problem of testing return predictability in the presence of highly persistent predictors and stationary exogenous covariates. The exogenous covariate may be a macroeconomic, environmental, or other predetermined state variable whose effect on returns is not well described by a constant intercept. The data-generating process (DGP) is specified by the following system of equations:

1. **Predictive Regression:**
$$Y_t = m(W_{t-1}) + \beta X_{t-1} + U_t \tag{1}$$
$$X_t = \theta + \rho X_{t-1} + \varepsilon_t \tag{2}$$

2. **Endogeneity:**
$$U_t = \phi \varepsilon_t + z_t \quad \text{with } E[z_t \mid \varepsilon_t] = 0 \tag{3}$$

where Y_t represents asset returns, W_{t-1} is a stationary exogenous covariate, and X_{t-1} is a persistent financial predictor. The function $m(W_{t-1})$ is an unknown nonparametric nuisance component: it can be viewed as a flexible intercept or exogenous-control term that absorbs the direct effect of W_{t-1} on returns. The key timing restriction is contemporaneous sieve orthogonality between the exogenous covariate and the persistent predictor weight at the sampling frequency used for inference. This restriction does not rule out effects of W_{t-1} on returns through $m(W_{t-1})$, nor does it rule out adjustment in future values of X_t.

Throughout the theoretical development, W_{t-1} should be read as a generic stationary exogenous covariate. The weather-shock interpretation used below is one empirical application of this more general setup.

Motivated by Perron (1989), our goal is to investigate return predictability while avoiding size distortions or false inferences caused by near unit roots in the predictor dynamics (i.e., $\rho \approx 1$). This is achieved by testing the null hypothesis of no predictability:

$$\begin{aligned} H_0 &: \quad \beta = 0 \\ H_1 &: \quad \beta \neq 0 \end{aligned} \tag{4}$$

By extending the empirical likelihood (EL) framework of Cai et al. (2026), we propose a two-stage Orthogonalized Sieve-EL framework that achieves a standard χ_1^2 limiting distribution under the null.

2 Empirical Interpretation: Weather Shocks and Cocoa Futures

The framework can be applied to agricultural commodity markets by interpreting W_{t-1} as a local weather variable, such as precipitation or average temperature, and Y_t as excess returns on cocoa futures. The 2023–2024 cocoa market provides a useful example: severe weather conditions in West Africa contributed to supply deficits and large price movements, while lagged prices and related financial predictors remained highly persistent.

This empirical setting fits the general framework for three reasons. First, weather variables are plausibly exogenous to contemporaneous financial-market innovations at the daily sampling frequency, while still being allowed to affect subsequent returns through the structural component $m(W_{t-1})$. Second, the relation between weather and supply is nonlinear, so a flexible function $m(W_{t-1})$ is more appropriate than a constant intercept or a linear control. Third, financial predictors such as lagged prices can be highly persistent, so inference on β must remain valid across near-unit-root persistence regimes.

Thus, in the cocoa application, the generic nuisance component $m(W_{t-1})$ has the interpretation of a nonlinear weather-response curve. The econometric method itself, however, does not depend on weather specifically; it only requires a stationary exogenous covariate satisfying the orthogonality and regularity conditions stated below.

3 The Orthogonalized Sieve-EL Framework

We handle the unknown nonparametric nuisance component $m(W_{t-1})$ by approximating it using a sieve basis of dimension K (e.g., polynomials or splines). Let $\boldsymbol{P}_K(W_{t-1}) = (p_1(W_{t-1}), \ldots, p_K(W_{t-1}))'$ be the $K \times 1$ vector of these basis functions.

3.1 The Step-by-Step Algorithm

The estimation and inference procedure is implemented sequentially via the following three steps. The key idea is to profile out the flexible intercept component and then construct an EL moment that is orthogonal to the same sieve space.

Step 1: Profiling out the Nonparametric Nuisance Component

For each candidate parameter β under the null hypothesis, we profile out the unknown function $m(W_{t-1})$, which plays the role of a flexible intercept or exogenous-control component. Let $\boldsymbol{Y} = (Y_1, \ldots, Y_T)'$ and $\boldsymbol{X}_{lag} = (X_0, \ldots, X_{T-1})'$. Let $\boldsymbol{P}$ be the $T \times K$ matrix whose t-th row is $\boldsymbol{P}_K(W_{t-1})'$. The profile sieve coefficients for the nuisance component are:

$$\widehat{\boldsymbol{c}}_K(\beta) = (\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}'(\boldsymbol{Y} - \beta \boldsymbol{X}_{lag}) \tag{5}$$

The profile residual is defined using the sieve annihilator matrix $\boldsymbol{M}_K = \boldsymbol{I}_T - \boldsymbol{P}(\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}'$:

$$\widehat{\boldsymbol{u}}(\beta) = \boldsymbol{M}_K(\boldsymbol{Y} - \beta \boldsymbol{X}_{lag}) \tag{6}$$

Under the null hypothesis $\beta = \beta_0$, substituting the true DGP $\boldsymbol{Y} = \boldsymbol{P}\boldsymbol{c} + \boldsymbol{r} + \beta_0 \boldsymbol{X}_{lag} + \boldsymbol{U}$ (where $\boldsymbol{r}$ is the approximation error) yields:

$$\widehat{\boldsymbol{u}}(\beta_0) = \boldsymbol{M}_K(\boldsymbol{P}\boldsymbol{c} + \boldsymbol{r} + \boldsymbol{U}) = \boldsymbol{M}_K(\boldsymbol{r} + \boldsymbol{U}) \tag{7}$$

Step 2: The Sieve Weight Projection

In standard Empirical Likelihood, estimating a high-dimensional nuisance component can affect the limiting distribution. To neutralize this generated-nuisance effect, we construct an orthogonalized weight function. Let $\boldsymbol{w} = (w_1, \ldots, w_T)'$ where $w_t = w(X_{t-1})$ is the raw weight (e.g., $\tanh(X_{t-1}/c_w)$). We project the weights onto the same sieve basis and define the orthogonalized weight $\boldsymbol{w}^c$ as the residual:

$$\boldsymbol{w}^c = \boldsymbol{M}_K \boldsymbol{w} \tag{8}$$

Let $w_t^c = (\boldsymbol{w}^c)_t$. By the properties of orthogonal projection, the orthogonalized weights are exactly orthogonal to the sieve basis in-sample:

$$\sum_{t=1}^{T} \boldsymbol{P}_K(W_{t-1}) w_t^c \equiv \boldsymbol{0} \tag{9}$$

Step 3: Algebraic Elimination in the EL Moment Condition

For testing $H_0 : \beta = \beta_0$, the EL moment condition is evaluated as:

$$\widehat{Z}_t(\beta_0) = \widehat{u}_t(\beta_0) w_t^c \tag{10}$$

Summing the moment conditions across the sample to evaluate the numerator of the EL statistic:

$$\sum_{t=1}^{T} \widehat{Z}_t(\beta_0) = \widehat{\boldsymbol{u}}(\beta_0)' \boldsymbol{w}^c = (\boldsymbol{r} + \boldsymbol{U})' \boldsymbol{M}_K \boldsymbol{M}_K \boldsymbol{w} = (\boldsymbol{r} + \boldsymbol{U})' \boldsymbol{w}^c \tag{11}$$

The fitted sieve component is thereby eliminated algebraically. The remaining approximation residual is controlled by the sieve rate conditions, leaving:

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{T} \widehat{Z}_t(\beta_0) = \frac{1}{\sqrt{T}} \sum_{t=1}^{T} U_t w_t^c + \underbrace{\frac{1}{\sqrt{T}} \sum_{t=1}^{T} r_t w_t^c}_{= o_p(1)} \tag{12}$$

Because the moment condition is orthogonal to the fitted nuisance space, the sample variance of $\widehat{Z}_t(\beta_0)$ consistently estimates the relevant innovation variance under the stated rate conditions. As a result, the Empirical Likelihood ratio statistic (where $\widehat{\lambda}$ is the Lagrange multiplier) achieves the standard chi-squared limit:

$$\ell_T(\beta_0) = 2 \sum_{t=1}^{T} \log(1 + \widehat{\lambda} \widehat{Z}_t(\beta_0)) \xrightarrow{d} \chi_1^2 \tag{13}$$

3.2 Comparison with Cai et al. (2026)

Our framework differs from the approach of Cai et al. (2026) in two fundamental aspects:

- **Nuisance Dimensionality**: Cai et al. (2026) handle a fixed scalar intercept α, whereas our model contains an infinite-dimensional nuisance function $m(W_{t-1})$, which requires a sieve approximation of dimension $K \to \infty$.
- **Orthogonalization Space**: The true generalization is that the demeaning against a constant 1 is replaced by residualizing against the sieve basis $\boldsymbol{P}_K(W)$. In Cai et al. (2026), the weight centering $w^c(x) = w(x) - \frac{1}{T} \sum w(X_s)$ represents a projection onto the constant subspace span$\{\mathbf{1}\}$. In contrast, our orthogonalized weight $\boldsymbol{w}^c$ is projected onto the K-dimensional sieve space span$\{\boldsymbol{P}_K(W_{t-1})\}$. We keep the AR(1) equation as the persistence DGP, but we do not use AR(1) OLS residuals in the EL statistic.

4 The Growth Rate of the Sieve Dimension K

To establish the asymptotic validity of our test statistic, we must place bounds on the growth rate of the tuning parameter K relative to the sample size T.

4.1 Lower Bound (Bias Elimination)

Assume the true nuisance function $m(W)$ has smoothness s (i.e., is s times continuously differentiable). Sieve theory implies that the approximation error r_t satisfies $\sup |r_t| = O_p(K^{-s})$. To ensure that the approximation bias does not distort the asymptotic distribution, we require $T^{-1/2} \sum_{t=1}^{T} r_t w_t^c = o_p(1)$, which translates to:

$$\sqrt{T} K^{-s} \to 0 \quad \Longrightarrow \quad K \gg T^{\frac{1}{2s}} \tag{14}$$

4.2 Upper Bound (Variance Control and Leakage Mitigation)

Estimating K nuisance parameters introduces in-sample variance inflation and potential projection leakage. Under mixing conditions for the exogenous covariate process W_t, this leakage does not accumulate, and the leading variance penalty scales as $K/\sqrt{T}$. To prevent variance inflation from affecting the EL ratio, we require:

$$\frac{K}{\sqrt{T}} \to 0 \quad \implies \quad K \ll T^{\frac{1}{2}} \tag{15}$$

4.3 Growth rate of K

Combining these bounds yields the formal asymptotic growth condition:

$$T^{\frac{1}{2s}} \ll K \ll T^{\frac{1}{2}} \tag{16}$$

If we assume $m(W)$ is twice-differentiable ($s = 2$), the condition becomes $T^{1/4} \ll K \ll T^{1/2}$. In practice, setting $K \approx T^{1/3}$ satisfies these bounds, ensuring that both approximation bias and variance inflation vanish asymptotically.

5 Formal Orthogonality Assumptions

To ensure the algebraic orthogonalization rigorously eliminates the fitted nonparametric nuisance component, we must state the relationship between the persistent financial predictor and the stationary exogenous covariate. We impose the following two conditions on W_{t-1}:

Assumption 5.1 (Strict Exogeneity to the Market Innovation) *The exogenous covariate W_{t-1} is strictly exogenous to the financial market innovation U_t. Mathematically:*

$$\mathbb{E}[U_t \mid \mathcal{F}_{t-1}, W_{t-1}] = 0. \tag{17}$$

Remark. This condition says the exogenous covariate is not a proxy for the residual market innovation U_t after conditioning on predetermined market information. It still allows W_{t-1} to affect returns through the structural component $m(W_{t-1})$ and to be incorporated into future predictor dynamics.

Assumption 5.2 (Contemporaneous Sieve Orthogonality) *At the sampling frequency used for inference, the exogenous covariate is contemporaneously orthogonal to the predetermined persistent predictor in the sieve directions used for profiling. For the EL weight $w(X_{t-1})$,*

$$\mathbb{E}[w(X_{t-1}) \mid W_{t-1}] = \mathbb{E}[w(X_{t-1})]. \tag{18}$$

Equivalently, for any centered nonconstant sieve vector $\boldsymbol{R}_K(W_{t-1})$,

$$\mathbb{E}\left[\boldsymbol{R}_K(W_{t-1})\{w(X_{t-1}) - \mathbb{E}w(X_{t-1})\}\right] = \mathbf{0}. \tag{19}$$

This is a contemporaneous timing restriction, not a dynamic exclusion: W_{t-1} may affect Y_t through $m(W_{t-1})$ and may affect subsequent returns and future predictor values.

Remark. This assumption is the population version of the sample orthogonality condition used in the proof, $\|T^{-1}\boldsymbol{R}'\widetilde{\boldsymbol{w}}\| = O_p(\sqrt{K/T})$. It preserves the persistent predictor signal at the moment used for inference while allowing the exogenous covariate to enter later market adjustment.

6 Wilks' Theorem for Mean Predictability

This section gives a formal Wilks theorem for the mean-predictability test. It is useful to write the nuisance step in null-imposed profile form. Let

$$\boldsymbol{Y} = (Y_1, \ldots, Y_T)', \qquad \boldsymbol{X}_{lag} = (X_0, \ldots, X_{T-1})',$$

and let $\boldsymbol{P}$ be the $T \times K$ matrix whose t-th row is $\boldsymbol{P}_K(W_{t-1})'$. Define

$$\boldsymbol{\Pi}_K = \boldsymbol{P}(\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}', \qquad \boldsymbol{M}_K = \boldsymbol{I}_T - \boldsymbol{\Pi}_K.$$

For each candidate value β, define the null-imposed profile sieve coefficient and the corresponding residual by

$$\widehat{\boldsymbol{c}}_K(\beta) = (\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}'(\boldsymbol{Y} - \beta\boldsymbol{X}_{lag}), \qquad \widehat{\boldsymbol{u}}(\beta) = \boldsymbol{M}_K(\boldsymbol{Y} - \beta\boldsymbol{X}_{lag}). \tag{20}$$

Let $w_t = w(X_{t-1})$, $\boldsymbol{w} = (w_1, \ldots, w_T)'$, and define the sieve-orthogonalized weight

$$\boldsymbol{w}^c = \boldsymbol{M}_K\boldsymbol{w}, \qquad w_t^c = (\boldsymbol{w}^c)_t. \tag{21}$$

The scalar estimating equation is

$$\widehat{Z}_t(\beta) = \widehat{u}_t(\beta)w_t^c. \tag{22}$$

The empirical likelihood and its log-likelihood ratio are

$$L_T(\beta) = \sup\left\{\prod_{t=1}^{T}(T\pi_t) : \pi_t \geq 0, \quad \sum_{t=1}^{T}\pi_t = 1, \quad \sum_{t=1}^{T}\pi_t\widehat{Z}_t(\beta) = 0\right\}, \tag{23}$$

$$\ell_T(\beta) = -2\log L_T(\beta) = 2\sum_{t=1}^{T}\log\{1 + \widehat{\lambda}(\beta)\widehat{Z}_t(\beta)\}, \tag{24}$$

where $\widehat{\lambda}(\beta)$ solves

$$\sum_{t=1}^{T}\frac{\widehat{Z}_t(\beta)}{1 + \widehat{\lambda}(\beta)\widehat{Z}_t(\beta)} = 0. \tag{25}$$

For a vector $\boldsymbol{a} = (a_1, \ldots, a_T)'$, write $\|\boldsymbol{a}\|_T^2 = T^{-1}\sum_{t=1}^{T} a_t^2$. Also write

$$\overline{w}_T = T^{-1}\sum_{t=1}^{T} w_t, \qquad \widetilde{w}_t = w_t - \overline{w}_T.$$

Assumption 1 (CCHT mean-score conditions)**.** *The predictor satisfies* $X_t = \theta + \rho_T X_{t-1} + \varepsilon_t$, *where* ρ_T *belongs to one of the persistence classes considered by Cai, Chen, Hong, and Tsvetanov (CCHT), including the stationary, mildly integrated, local-to-unity, integrated, and mildly explosive cases. With respect to the filtration*

$$\mathcal{H}_t = \sigma\{(z_s, \varepsilon_s) : s \leq t\} \vee \sigma\{W_s : s \leq t\},$$

$E(U_t \mid \mathcal{H}_{t-1}) = 0$, $\sup_t E|U_t|^{2+\delta} < \infty$ *for some* $\delta > 0$, *and the conditional-variance and weight conditions used for the CCHT mean theorem hold. In particular, for*

$$A_T = \frac{1}{\sqrt{T}}\sum_{t=1}^{T} U_t\widetilde{w}_t, \qquad V_T = \frac{1}{T}\sum_{t=1}^{T} U_t^2\widetilde{w}_t^2,$$

there exists an a.s. positive, possibly random, variable η^2 *such that*

$$A_T \xrightarrow{\text{st}} MN(0, \eta^2), \qquad V_T \xrightarrow{p} \eta^2, \qquad \max_{1\leq t\leq T}|U_t\widetilde{w}_t| = o_p(\sqrt{T}). \tag{26}$$

The raw weight is uniformly bounded and satisfies the CCHT saturation condition.

Assumption 2 (Exogenous-covariate sieve orthogonality and projection)**.** *The first component of* $\boldsymbol{P}_K(W)$ *is one. After centering the remaining basis functions in sample, write the corresponding* $T \times (K-1)$ *matrix as* $\boldsymbol{R}$, *so that* $\boldsymbol{1}_T'\boldsymbol{R} = \boldsymbol{0}$ *and* $\operatorname{span}(\boldsymbol{P}) = \operatorname{span}(\boldsymbol{1}_T, \boldsymbol{R})$. *The primitive timing restriction is contemporaneous sieve orthogonality:*

$$\mathbb{E}\left[\boldsymbol{R}_K(W_{t-1})\{w(X_{t-1}) - \mathbb{E}w(X_{t-1})\}\right] = \boldsymbol{0},$$

where $\boldsymbol{R}_K(W_{t-1})$ denotes the population counterpart of the centered nonconstant sieve vector. This condition allows W_{t-1} to affect Y_t and future values of X; it only rules out contemporaneous projection of the predictor weight onto the exogenous-covariate sieve. The strict-exogeneity condition in Assumption 1 holds with contemporaneous exogenous-covariate information included in the conditioning set.

There are constants $0 < c < C < \infty$ such that, with probability tending to one,

$$c \leq \lambda_{\min}(T^{-1}\boldsymbol{R}'\boldsymbol{R}) \leq \lambda_{\max}(T^{-1}\boldsymbol{R}'\boldsymbol{R}) \leq C.$$

Let $\zeta_K = \max_{t\leq T}\|\boldsymbol{R}_t\|$, where $\boldsymbol{R}_t'$ is row t of $\boldsymbol{R}$. The sample analogue of the population orthogonality and the remaining increasing-dimensional score bounds are:

$$\left\|T^{-1}\boldsymbol{R}'\widetilde{\boldsymbol{w}}\right\| = O_p\left(\sqrt{K/T}\right), \tag{27}$$

$$\left\|T^{-1/2}\boldsymbol{R}'\boldsymbol{U}\right\| = O_p(\sqrt{K}), \tag{28}$$

$$\left\|T^{-1}\boldsymbol{P}'\boldsymbol{U}\right\| = O_p\left(\sqrt{K/T}\right), \tag{29}$$

where $\widetilde{\boldsymbol{w}} = (\widetilde{w}_1, \ldots, \widetilde{w}_T)'$ and $\boldsymbol{U} = (U_1, \ldots, U_T)'$. These bounds follow under the stated sieve orthogonality, exponential mixing of W_t, uniformly bounded basis moments, and the martingale moment condition in Assumption 1.

Assumption 3 (Approximation and growth rates)**.** *For some $\boldsymbol{c}_K$,*

$$m(W_{t-1}) = \boldsymbol{P}_K(W_{t-1})'\boldsymbol{c}_K + r_{K,t},$$

where

$$\|\boldsymbol{r}_K\|_T = O(a_K), \qquad \|\boldsymbol{r}_K\|_\infty = O(a_K), \qquad \sqrt{T}\,a_K \longrightarrow 0.$$

The sieve projector is stable in sup norm on the approximation residual, so that $\|\boldsymbol{M}_K\boldsymbol{r}_K\|_\infty = O(a_K)$. Moreover,

$$K/\sqrt{T} \longrightarrow 0, \qquad \zeta_K\sqrt{K/T} \longrightarrow 0. \tag{30}$$

For normalized spline bases, for which typically $\zeta_K \asymp \sqrt{K}$, the second condition in (30) is implied by $K/\sqrt{T} \to 0$. If $a_K \asymp K^{-s/d}$, a sufficient rate window is $T^{d/(2s)} \ll K \ll T^{1/2}$, which is nonempty when $s > d$.

Assumption 4 (Nondegeneracy and convex hull)**.** *There exists $\underline{\eta} > 0$ such that $P(\eta^2 \geq \underline{\eta}^2) = 1$. In addition,*

$$P\left\{\min_{1\leq t\leq T}\widehat{Z}_t(\beta_0) < 0 < \max_{1\leq t\leq T}\widehat{Z}_t(\beta_0)\right\} \longrightarrow 1.$$

Lemma 1 (Negligibility of the growing sieve projection)**.** *Under Assumptions 2–3,*

$$\|\boldsymbol{w}^c - \widetilde{\boldsymbol{w}}\|_T = O_p\left(\sqrt{K/T}\right), \tag{31}$$

$$\max_{t\leq T}|w_t^c - \widetilde{w}_t| = O_p\left(\zeta_K\sqrt{K/T}\right) = o_p(1), \tag{32}$$

$$\frac{1}{\sqrt{T}}\sum_{t=1}^{T}U_t(w_t^c - \widetilde{w}_t) = O_p(K/\sqrt{T}) = o_p(1), \tag{33}$$

$$\|\boldsymbol{\Pi}_K\boldsymbol{U}\|_T = O_p\left(\sqrt{K/T}\right), \tag{34}$$

$$\max_{t\leq T}|(\boldsymbol{\Pi}_K\boldsymbol{U})_t| = O_p\left(\zeta_K\sqrt{K/T}\right) = o_p(1). \tag{35}$$

Proof. Because the sieve contains a constant and $\boldsymbol{1}_T'\boldsymbol{R} = \boldsymbol{0}$, residualizing $\boldsymbol{w}$ on $\boldsymbol{P}$ is equivalent to residualizing $\widetilde{\boldsymbol{w}}$ on $\boldsymbol{R}$. Hence

$$\boldsymbol{w}^c = \widetilde{\boldsymbol{w}} - \boldsymbol{R}\widehat{\boldsymbol{\delta}}_K, \qquad \widehat{\boldsymbol{\delta}}_K = (\boldsymbol{R}'\boldsymbol{R})^{-1}\boldsymbol{R}'\widetilde{\boldsymbol{w}}.$$

The Gram-matrix condition and (27) imply $\|\widehat{\boldsymbol{\delta}}_K\| = O_p(\sqrt{K/T})$. Therefore,

$$\|\boldsymbol{R}\widehat{\boldsymbol{\delta}}_K\|_T^2 = \widehat{\boldsymbol{\delta}}_K'(T^{-1}\boldsymbol{R}'\boldsymbol{R})\widehat{\boldsymbol{\delta}}_K = O_p(K/T),$$

which proves (31); and

$$\max_t |\boldsymbol{R}_t'\widehat{\boldsymbol{\delta}}_K| \le \zeta_K \|\widehat{\boldsymbol{\delta}}_K\| = O_p\Big(\zeta_K\sqrt{K/T}\Big),$$

which proves (32). Moreover,

$$\frac{1}{\sqrt{T}}\boldsymbol{U}'(\boldsymbol{w}^c - \widetilde{\boldsymbol{w}}) = -\Big(T^{-1/2}\boldsymbol{R}'\boldsymbol{U}\Big)'\widehat{\boldsymbol{\delta}}_K = O_p(\sqrt{K})O_p\Big(\sqrt{K/T}\Big) = O_p(K/\sqrt{T}),$$

giving (33).

Finally, let $\widehat{\boldsymbol{a}}_U = (\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}'\boldsymbol{U}$. The Gram condition and (29) imply $\|\widehat{\boldsymbol{a}}_U\| = O_p(\sqrt{K/T})$. The same quadratic-form and maximum-row arguments give (34) and (35). $\square$

Lemma 2 (Asymptotic equivalence of the feasible and oracle scores)**.** *Under Assumptions 1–3, under $H_0 : \beta = \beta_0$,*

$$S_T := \frac{1}{\sqrt{T}}\sum_{t=1}^T \widehat{Z}_t(\beta_0) = \frac{1}{\sqrt{T}}\sum_{t=1}^T U_t\widetilde{w}_t + o_p(1), \tag{36}$$

$$\widehat{V}_T := \frac{1}{T}\sum_{t=1}^T \widehat{Z}_t(\beta_0)^2 = \frac{1}{T}\sum_{t=1}^T U_t^2\widetilde{w}_t^2 + o_p(1), \tag{37}$$

$$\max_{t\le T}|\widehat{Z}_t(\beta_0)| = o_p(\sqrt{T}). \tag{38}$$

Proof. Under the null,

$$\boldsymbol{Y} - \beta_0\boldsymbol{X}_{lag} = \boldsymbol{P}\boldsymbol{c}_K + \boldsymbol{r}_K + \boldsymbol{U}, \qquad \widehat{\boldsymbol{u}}(\beta_0) = \boldsymbol{M}_K(\boldsymbol{U} + \boldsymbol{r}_K).$$

Because $\boldsymbol{M}_K$ is symmetric and idempotent and $\boldsymbol{w}^c = \boldsymbol{M}_K\boldsymbol{w}$,

$$\begin{aligned} S_T &= T^{-1/2}\widehat{\boldsymbol{u}}(\beta_0)'\boldsymbol{w}^c \\ &= T^{-1/2}(\boldsymbol{U} + \boldsymbol{r}_K)'\boldsymbol{M}_K^2\boldsymbol{w} \\ &= T^{-1/2}(\boldsymbol{U} + \boldsymbol{r}_K)'\boldsymbol{w}^c. \end{aligned}$$

Lemma 1 gives $T^{-1/2}\boldsymbol{U}'\boldsymbol{w}^c = T^{-1/2}\boldsymbol{U}'\widetilde{\boldsymbol{w}} + o_p(1)$. Also, by Cauchy–Schwarz and the contraction property of an orthogonal projection,

$$\Big|T^{-1/2}\boldsymbol{r}_K'\boldsymbol{w}^c\Big| \le \sqrt{T}\,\|\boldsymbol{r}_K\|_T\|\boldsymbol{w}^c\|_T \le \sqrt{T}\,\|\boldsymbol{r}_K\|_T\|\boldsymbol{w}\|_T = o_p(1).$$

This proves (36).

For the second-moment result, define

$$e_{u,t} = \widehat{u}_t(\beta_0) - U_t, \qquad e_{w,t} = w_t^c - \widetilde{w}_t.$$

Then

$$\boldsymbol{e}_u = -\boldsymbol{\Pi}_K\boldsymbol{U} + \boldsymbol{M}_K\boldsymbol{r}_K,$$

so that, by Lemma 1 and projection contraction,

$$\|\boldsymbol{e}_u\|_T = O_p\Big(\sqrt{K/T} + a_K\Big) = o_p(1).$$

Moreover, $\max_t|e_{w,t}| = o_p(1)$, $\max_t|w_t^c| = O_p(1)$, and $T^{-1}\sum_t U_t^2 = O_p(1)$. Since

$$\widehat{Z}_t(\beta_0) - U_t\widetilde{w}_t = e_{u,t}w_t^c + U_te_{w,t},$$

it follows that

$$\frac{1}{T}\sum_{t=1}^{T}\{\widehat{Z}_t(\beta_0)-U_t\widetilde{w}_t\}^2 \le 2\max_t |w_t^c|^2\|\boldsymbol{e}_u\|_T^2 + 2\max_t |e_{w,t}|^2\frac{1}{T}\sum_{t=1}^{T}U_t^2$$
$$= o_p(1).$$

Equation (37) now follows from Cauchy–Schwarz.

Finally, $\max_t |U_t| = o_p(\sqrt{T})$ follows from the uniform $(2+\delta)$-moment bound. Together with (35), the sup-norm stability of $\boldsymbol{M}_K\boldsymbol{r}_K$, and $\max_t |w_t^c| = O_p(1)$, this proves (38). □

Theorem 1 (Wilks phenomenon for sieve-orthogonalized mean EL)**.** *Suppose Assumptions 1–4 hold. Then, under* $H_0 : \beta = \beta_0$*,*

$$\ell_T(\beta_0) \xrightarrow{d} \chi_1^2. \tag{39}$$

The conclusion holds for each predictor-persistence class covered by the CCHT mean theorem.

Proof. By Lemma 2 and Assumption 1,

$$S_T \xrightarrow{\text{st}} MN(0,\eta^2), \qquad \widehat{V}_T \xrightarrow{p} \eta^2.$$

Consequently,

$$\frac{S_T}{\sqrt{\widehat{V}_T}} \xrightarrow{\text{st}} N(0,1), \qquad \frac{S_T^2}{\widehat{V}_T} \xrightarrow{d} \chi_1^2. \tag{40}$$

It remains to connect the self-normalized statistic to empirical likelihood. Write $\widehat{Z}_t = \widehat{Z}_t(\beta_0)$ and $\widehat{\lambda} = \widehat{\lambda}(\beta_0)$. The convex-hull condition, $S_T = O_p(1)$, $\widehat{V}_T \to_p \eta^2 > 0$, and $\max_t |\widehat{Z}_t| = o_p(\sqrt{T})$ imply the standard scalar-EL bounds

$$\widehat{\lambda} = O_p(T^{-1/2}), \qquad \max_{t\le T} |\widehat{\lambda}\widehat{Z}_t| = o_p(1).$$

Expanding the multiplier equation (25) gives

$$0 = \sum_{t=1}^{T}\widehat{Z}_t - \widehat{\lambda}\sum_{t=1}^{T}\widehat{Z}_t^2 + o_p(\sqrt{T}),$$

and hence

$$\widehat{\lambda} = \frac{\sum_{t=1}^{T}\widehat{Z}_t}{\sum_{t=1}^{T}\widehat{Z}_t^2} + o_p(T^{-1/2}). \tag{41}$$

A second-order expansion of (24), together with $\max_t |\widehat{\lambda}\widehat{Z}_t| = o_p(1)$, yields

$$\ell_T(\beta_0) = 2\widehat{\lambda}\sum_{t=1}^{T}\widehat{Z}_t - \widehat{\lambda}^2\sum_{t=1}^{T}\widehat{Z}_t^2 + o_p(1)$$
$$= \frac{\left(T^{-1/2}\sum_{t=1}^{T}\widehat{Z}_t\right)^2}{T^{-1}\sum_{t=1}^{T}\widehat{Z}_t^2} + o_p(1)$$
$$= \frac{S_T^2}{\widehat{V}_T} + o_p(1).$$

The result follows from (40) and Slutsky's theorem. □

Corollary 1 (Test of no mean predictability)**.** *For* $H_0 : \beta = 0$*, reject at asymptotic level* α *whenever*

$$\ell_T(0) > \chi^2_{1,1-\alpha}.$$

Equivalently, the asymptotic p-value is $1 - F_{\chi_1^2}\{\ell_T(0)\}$*. A confidence set for* β *is obtained by inversion:*

$$\mathcal{C}_{1-\alpha} = \{\beta : \ell_T(\beta) \le \chi^2_{1,1-\alpha}\}.$$

Remark 1 (Relation to the unrestricted two-stage estimator)**.** *The profile construction* (20) *is the cleanest route to Theorem 1: at* β_0*, the fitted nuisance error is exactly* $\mathbf{\Pi}_K(\boldsymbol{U} + \boldsymbol{r}_K)$*, whose prediction norm is* $O_p(\sqrt{K/T} + a_K)$*. If one instead retains a separately estimated unrestricted coefficient* $\widehat{\mathbf{c}}$*, the same theorem continues to hold provided one proves the additional generated-nuisance condition*

$$\frac{1}{T}\sum_{t=1}^{T}\left\{\boldsymbol{P}_K(W_{t-1})'(\widehat{\boldsymbol{c}} - \boldsymbol{c}_K)\right\}^2 (w_t^c)^2 = o_p(1)$$

and the corresponding maximum-term bound. Those statements do not follow from the algebraic identity $\boldsymbol{P}'\boldsymbol{w}^c = \mathbf{0}$ *alone.*

Remark 2 (Why the same argument is not immediately a quantile theorem)**.** *For a quantile score* $\psi_\tau(u) = \tau - \mathbf{1}\{u < 0\}$*, the nuisance error enters nonlinearly:*

$$\psi_\tau(U_{t,\tau} + r_{K,t} - \boldsymbol{P}_K(W_{t-1})'(\widehat{\boldsymbol{c}}_\tau - \boldsymbol{c}_{K,\tau}))\,.$$

Therefore $\boldsymbol{P}'\boldsymbol{w}^c = \mathbf{0}$ *does not algebraically cancel the first-stage error. A first-order expansion instead produces a density-weighted term of the form*

$$\sum_{t=1}^{T} f_{t,\tau}(0)\boldsymbol{P}_K(W_{t-1})w_t^c,$$

which is not generally zero. A quantile extension would require either a restrictive condition making $f_{t,\tau}(0)$ *constant with respect to the predictor, or a new density-weighted orthogonalization and additional nonparametric-rate arguments. Thus Theorem 1 is a complete and natural endpoint for the mean paper, whereas a general quantile extension is a separate methodological problem.*

7 Conclusion

Standard predictive regressions often absorb nonlinear exogenous state variables into a fixed intercept or the regression error. That simplification can leave important nuisance variation in the residual and can interact poorly with highly persistent predictors. This paper instead treats the intercept component as an unknown function $m(W)$, profiles it out under each candidate β, and residualizes the EL weight against the same sieve space.

The AR(1) specification is used only to describe the persistence class of the predictor. The empirical likelihood statistic itself does not estimate or include AR residuals. The resulting orthogonalized Sieve-EL statistic provides standard inference on mean predictability while allowing a flexible exogenous-control component. Weather shocks in commodity markets are one motivating application, but the theory applies more generally to stationary exogenous covariates satisfying the stated orthogonality and regularity conditions.